



Experimental Investigation of a Variable Collective Pitch Propeller at Incidence

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We extend our experiments on fixed-pitch propellers in forward flight at various incidence angles with respect to the freestream, to a propeller with variable collective pitch. Different combinations of blade pitch, incidence angle, and flight speed were tested at the University of Dayton Low-Speed Wind Tunnel in an open-jet test section. All six components of force and moment were measured, at incidence angles of -10 to 20 degrees. A higher thrust, power consumption, pitching moment, and yawing moment are measured at higher incidence angles, which agrees with the previous study and literature. We suggest curve fits to capture dependency on collective pitch, advance ratio, and incidence angle, for reduction to stability derivatives, and for prediction of trends beyond just the classical thrust/torque/power coefficients.

I. Nomenclature

C_T = Propeller Thrust Coefficient; $C_T = \frac{T}{\rho n^2 D^4}$	J = Propeller Advance Ratio; $J = \frac{V_\infty}{nD}$
C_{TA} = Propeller Axial Thrust Coefficient	n = Propeller Rotational Speed, (RPS)
C_P = Power Coefficient; $C_P = 2\pi C_Q = \frac{P}{\rho n^3 D^5}$	P = Power, (W)
C_Q = Propeller Torque Coefficient; $C_Q = \frac{Q}{\rho n^2 D^5}$	γ = Propeller Pitch, (m)
C_{QA} = Propeller Axial Torque Coefficient	Q = Torque, (Nm)
C_m = Propeller Pitching Moment Coefficient	R = Propeller Radius, (m)
C_n = Propeller Yawing Moment Coefficient	T = Thrust, (N)
D = Propeller Diameter, (m)	V_∞ = Freestream Velocity, (m/s)
FoM = Figure of Merit; $FoM = \frac{T \sqrt{\frac{T}{2\rho A}}}{P}$	ρ = Air Density, (kg/m ³)
	δ = Propeller Blade Pitch Angle, (deg)
	η = Propeller Efficiency, $\eta = \frac{c_T J}{c_P}$
	θ = Propeller-plane Incidence Angle, (deg)

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II. Introduction

The increasing interest in Advanced Air Mobility (AAM) broadens the usage of propeller-driven aerial vehicles, including unmanned aerial vehicles (UAVs) and electric Vertical take-off and Landing (eVTOL) air taxis. This expands the application scenario of the classical propeller from conventional fixed-wing aircraft to various forward flight conditions where the propeller disk is placed under different incidence angles, which is to say, applications more akin to a prop-rotor or helicopter rotor. The mainstream concept of a multi-copter, whether strictly rotor-borne or assisted by conventional wings, is predicated on some plurality of fixed-pitch propellers. The propeller rotational speed is varied for flight control and for prosecuting the various performance tasks (climb, forward/backward acceleration, steady level cruise, and so on). There aircraft tilts to fly forward, so that the lifting propellers – that is the only propellers – are now at some incidence angle.

The classical propeller experiment is for fixed-wing aircraft in forward flight, where the propeller sees an incidence angle. $\theta \sim 0^\circ$. References [1-7] are recent examples of fixed-pitch propellers. In general, at lower advance ratios ($J < \sim 0.2$), the thrust coefficient (C_T) and power coefficient (C_P) remain relatively constant, and as the advance ratio increases, both C_T and C_P fall as the propeller unloads, and eventually reaches zero at a sufficiently high J , depending on the propeller pitch (γ).

Propellers in higher incidence angles (non-axial flight) have also been investigated [4,5,8,9-11,14]. McCormick [6] notes that classical momentum theory is viable for both zero incidence (purely axial) and 90-degree incidence (purely edgewise) flight, concluding that if the approximation works for these two extremes, it should be applicable to all intermediate cases. Simmons and Hatke [4], McLemore and Cannon [5], Ribner [8], and Cai and Gunasekaran [14] investigated fixed-pitch propellers under a steady freestream at an incidence angle between 0 and 90 degrees. The propeller axial thrust coefficient (C_{TA}) and axial torque coefficient (C_{QA}) decrease with the increasing advance ratio at small θ , as expected. However, the C_{TA} and C_{QA} vs. J curve plateaus at higher incidence angles. C_{TA} and C_{QA} (seen in Figure 1) are nearly independent of J at $\theta \sim 60^\circ$, and show an increase with J at higher θ and approaching edgewise flight ($\theta = 90^\circ$). In other words, the propeller's ability to generate axial thrust increases when operating in edgewise flight. Meanwhile, due to the asymmetric blade loading at higher θ , higher pitching moment (C_m) and yawing moment (C_n) on the rotor are observed. A good correlation is found for these moments at different incidences, when plotting vs. the normal component of the advance ratio [4,11,14].

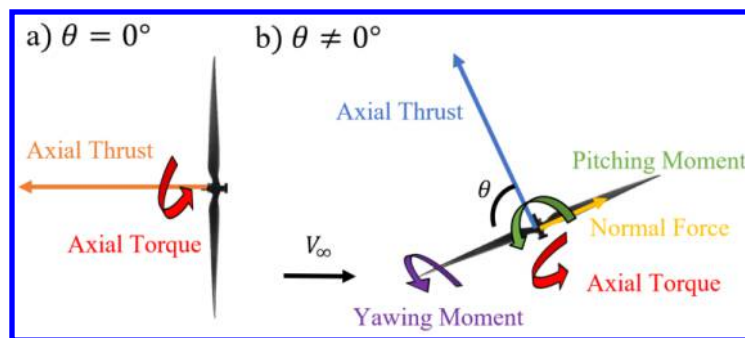


Figure 1. Schematic Showing Propeller Force and Moment Definition at a) $\theta = 0^\circ$ and b) $\theta \neq 0^\circ$

The study of a variable collective pitch propeller at nonzero incidence is more limited. Yaggy and Rogallo [11] conducted an experiment on a three-bladed rotor with variable pitch under $0^\circ \leq \theta \leq 85^\circ$. Besides the aforementioned findings, a significant difference in the rotor yawing moment (C_n) is found at the same θ under

the same incidence angle and different blade pitch angles (δ), while a smaller difference between the rotor pitching moment (C_m) is observed under the same conditions at different δ .

Our previous study [15] investigated forward thrust and vertical lift generation (Figure 1) for fixed-pitch propellers with different blade geometry at various incidence angles. At a small incidence angle, a propeller with a lower blade pitch distribution showed a higher efficiency in axial and forward thrust generation at lower forward flight advanced ratios; and vice versa at higher forward flight advanced ratios [1-3, 15]. This trend is preserved when increasing the propeller incidence angle until reaching nearly edgewise flight, where blade stall occurs on the propeller with a higher pitch-to-diameter ratio, especially at higher advance ratios [11,12]. Therefore, a propeller with a lower pitch may be more favorable for high-speed edgewise flight applications. Implementation of a variable pitch propeller potentially enables the propeller disk to operate at higher efficiency across the incidence angle range, from axial flight to edgewise flight ($0^\circ < \theta < 90^\circ$).

The goal of the current study is to extend Yaggy and Rogallo [11] to Reynolds numbers, parameters, and components relevant to modern drones, measuring all six components of force and moment. We consider collective pitch from the factory setting implied for a fixed-pitch propeller ($\delta = 0^\circ$) to past blade stall at a given advance ratio. The current manuscript focuses on small incidence angles, where the propeller does not far depart from the head-on axial flight. A follow-on study will consider incidence angles near, at, and beyond edgewise flight.

III. Experimental Setup

A) Data Acquisition

Experiments were conducted at the University of Dayton Low-Speed Wind Tunnel (UD-LSWT) in the open jet test section. The tunnel has an inlet contraction ratio of 16:1, a 762 mm x 762 mm testing section inlet, and a 1.37 m x 1.37 m collector. The turbulence intensity is 0.1%, as measured by a hot wire anemometer.

A sketch of the experimental setup is shown in Figure 2. Three blades taken from T-motor FA 16.2x5.3 propellers are paired with a T-motor V505 KV 260 brushless motor and a T-motor AT 115A 14S electronic speed controller [18]. The motor is powered by a GW Instek PSW 30-108, 1080W power supply. Each blade is retained in a bearing cage that reacts against centripetal force in an essentially rigid coupling. Variable collective pitch is accomplished via a single servo that drives a central pushrod through the motor, to a common linkage mechanism in the bearing-cage, and finally, a pitch-link at each blade grip. Conceptually the approach resembles a helicopter tail rotor assembly. There is no cyclic pitch variation. Blade collective pitch is statically calibrated, to reference measured blade angle to servo pulse width modulation (PWM) commands. Throttle settings are limited by the capacity of the power supply.

The propeller and motor assembly are attached to an ATI Industrial Automation Gamma six-component force/torque transducer [19]. The transducer has a maximum of 65N on its x and y-axis, 200N on its z-axis, and a maximum torque of 5 Nm for all axes. Thrust is the axial force along the z-axis of the sensor. The motor torque is the moment about the z-axis. This produces the classical torque coefficient, which when multiplied by rotational speed, gives the power coefficient, as usual. The positive y-axis of the sensor is also defined in Figure 2 and the orientation follows the right-hand rule for the positive x-axis.

The transducer is mounted on a 3D-printed interface, bolted to an 80-20 aluminum extrusion frame. The propeller can be tilted with respect to the tunnel freestream by changing the configuration of the aluminum frame. The entire assembly is attached to a 1.22 x 0.92 x 2.44-meter external aluminum frame to improve the rigidity of the testing stand and minimize the vibration. The external aluminum frame is secured to the lab floor, the inlet of the wind tunnel, and the ceiling of the wind tunnel plenum. A 10-mW laser and one Thorlabs PDA10A Photodiode

are used to measure the propeller rotational speed. A picture of the experimental setup is shown in Figure 3.

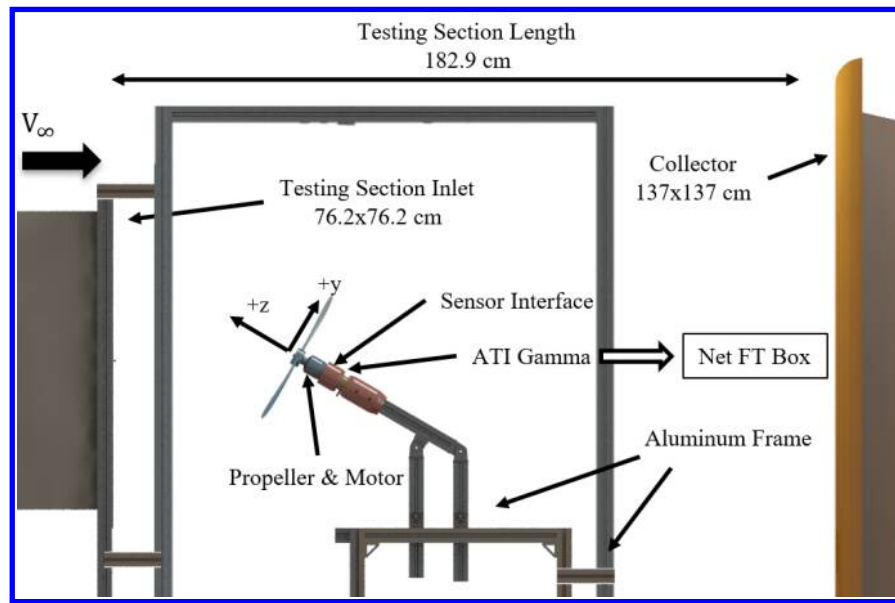


Figure 2. Schematic of the Experimental Setup at the UD-LSWT.

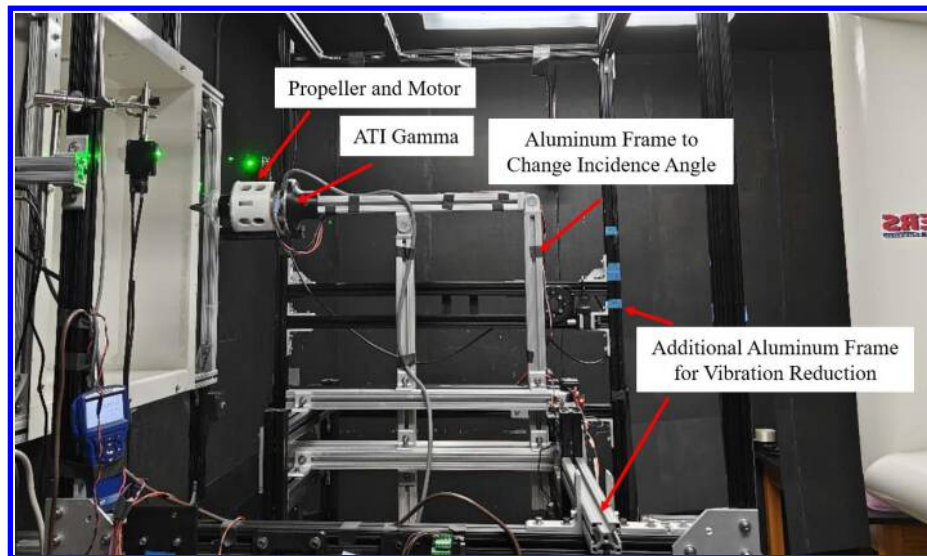


Figure 3. Experimental setup at the UD-LSWT.

Data was sampled at the rate of 1000 Hz for a 10-second duration at each data point. Two trials were run in each case to improve repeatability. To minimize sensor drift, tare values with the propeller powered off were taken before and after each test. The average of the two tares was subtracted from the respective thrust and torque readings. If the difference between the before and after tare value was greater than 0.2N of force on any axis, the experimental data was rejected. A Finite Impulse Response (FIR) band-stop filter [16] was applied to the data to filter out the vibration frequency caused by the propeller rotation, and the natural frequency of the aluminum test frame. Tap tests on the test stand with the propeller powered off, showed a structural natural frequency of $\sim 115\text{Hz}$. With the propeller thrust acting on the structure during the wind-on experiment, the structure does not show significant vibration up to the highest propeller rotational speed tested. The average uncertainty in the force measurement after filtration is $\pm 9.1\%$ across all advanced ratios tested, while the average uncertainty in torque is $\pm 7.9\%$. However, the results presented show excellent repeatability from both experimental runs.

B) Test Matrix

The test matrix is shown in Table 1. The propeller rotational speed varies between 50 and 105 RPS under different propeller blade pitch angles (δ). The peak Reynolds number is calculated at the 75% blade radius. The freestream velocity varies from 0 m/s to 30 m/s, resulting in an advance ratio, J , between 0 and 1.0 based on the propeller rotational speed. As the propeller will induce a different incoming freestream velocity at different blade pitch angles, for better comparison across cases, the advance ratio is calculated based on the calibrated wind tunnel speed without the influence of the propeller. Four different incidence angles are tested in the current study which represent the nearly axial flight condition. The blade collective pitch angle, which is measured based on the angle of the flat portion of each propeller blade at its grip varied from -12.0 to +20.8 degrees. $\delta = 0^\circ$ is the ‘factory default’ blade angle for a fixed-pitch propeller application. A positive blade pitch angle indicates the effective blade pitch distribution is higher than the ‘factory default’, and vice versa. The corresponding propeller root blade pitch angle vs. servo input PWM is shown in Figure 4 and is nearly linear, with a scatter of less than 0.5 degrees between different trails.

Table 1. Test Matrix

Propeller	Rotational Speed (RPS)	Peak Reynolds Number (x1000)	Advance Ratio	Incidence Angle (θ) (Deg)	Collective Pitch Angle (δ) (Deg)
T-motor FA 16.2x5.3	50 - 105	250	0 – 1.4	-10, 0, 10 20	-12 - 20.8

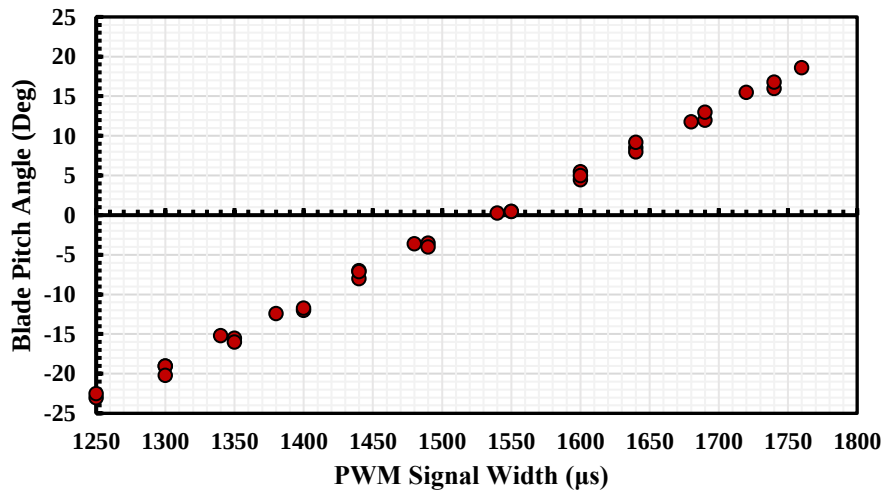


Figure 4. Departure in collective pitch from factory setting vs. PWM signal.

IV. Results

A. Hover Performance

C_{TA} and C_P vs. blade collective pitch angle (δ) in hover are shown in Figure 5, at three different throttle settings: 55%, 60 % and 65%. Propeller rotational speed varies with δ at constant PWM signal to the ESC, because the power-absorbed varies with δ . Results are grouped by the input throttle to the ESC. For $\delta = 0^\circ$, variation in the rotational speed between the minimum (55%) and the maximal throttle setting (65%) is ~22.5%; it is 17.5% at the highest δ tested. The results from different throttle settings showed good agreement, indicating no discernible Reynolds number effect or signal-to-noise effect within this range. C_{TA} increases with δ as expected; a linear relationship between C_T and δ is observed between $-10^\circ < \delta < 12.6^\circ$, with a slope of 0.0089 and an intercept of 0.1:

$$C_{T(\delta, J=0) Ideal} = 0.0089\delta + 0.1 \quad (1)$$

This is also represented by the red dashed line in Figure 5a. At a higher blade collective setting, stall occurs in hover, and C_T remains almost constant as δ increases. At around $\delta = -10^\circ$ there is a knee in the thrust curve, heading towards negative thrust, with a discernible slope change. This could speculatively be attributed to part of the propeller generating positive thrust, and part generating negative thrust. This is not strictly speaking stall, as the propeller continues to make negative aggregate thrust which increases (negatively) for a more negative collective angle. These settings are useful for a propeller used as a maneuvering thruster in hover but were not emphasized in the present study.

C_p also increases with δ at $\delta > -10^\circ$. Figure 6 shows the figure of merit for the cases in Figure 5. At $\delta < 0^\circ$, FOM falls rapidly, as evidently, the propeller is operating at blade effective angles of attack well below reasonable sectional lift-to-drag ratio. FOM reaches 0.56 at the factory setting ($\delta = 0^\circ$) and then climbs slowly, plateauing near 0.62 at $3^\circ < \delta < 12.6^\circ$, before falling again for $\delta > 12.6^\circ$, indicating the blade stall. Remembering that this propeller blade was developed for the fixed-pitch multi-copter industry, and thus, is intended for hover and for near-edgewise flight (at some inclination angle), one may conclude that the factory collective setting is reasonably close to “optimal” in hover, with allowances for edgewise flight. However, slightly better hover efficiency is possible if the collective setting is incrementally increased.

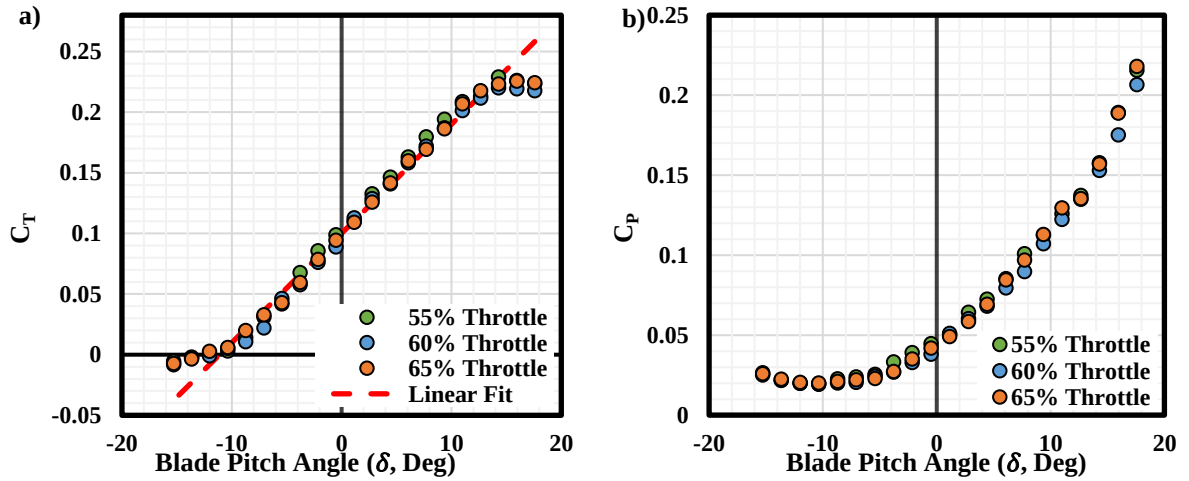


Figure 5. a) C_{TA} and b) C_P vs. collective pitch angle (δ) in hover.

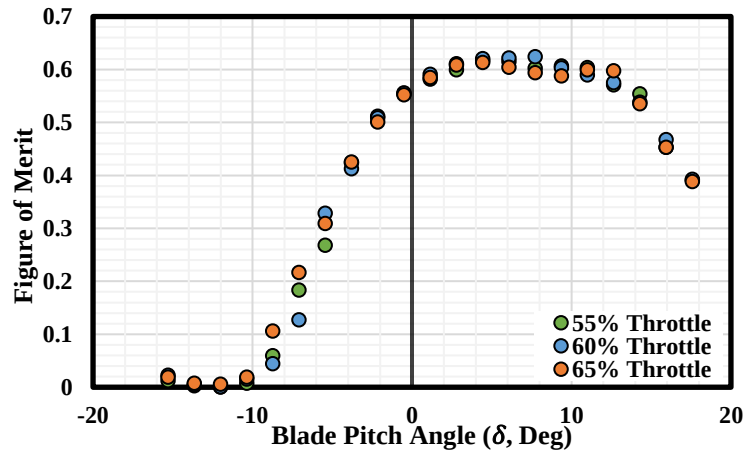


Figure 6. Figure of merit, for the cases in Figure 5.

B. Forward Flight at $\theta = 0^\circ$

As seen in the hover data, Reynolds number effects are relatively small for the rotational speed range tested. Therefore, results from one throttle setting are used for the forward flight analysis. Figure 7.a shows C_{TA} vs. J vs. δ at $\theta = 0^\circ$. Each legend represents a different δ from -0.5 to 20.8 degrees. Similarly to hover, a higher C_{TA} is measured at a higher δ under the same J . As expected, with increasing forward flight speed, C_{TA} reduces and eventually reaches zero, due to the reduction of the propeller blade's effective angle of attack [6,13]. There is no discernible rise in C_{TA} for the collective cases where the propeller is stalled in hover ($\delta = 15.9^\circ, 19.2^\circ, 20.8^\circ$). Perhaps this means that “stall” in hover was not complete, meaning that the collective setting would need to have been even higher, for full stall. C_P vs. J is shown in Figure 7.b. A similar trend as in hover is observed, where a higher C_{TA} results in a higher power consumption. C_P also falls with J , as the propeller unloads. The advanced ratio at which this happens depends of course on the collective angle.

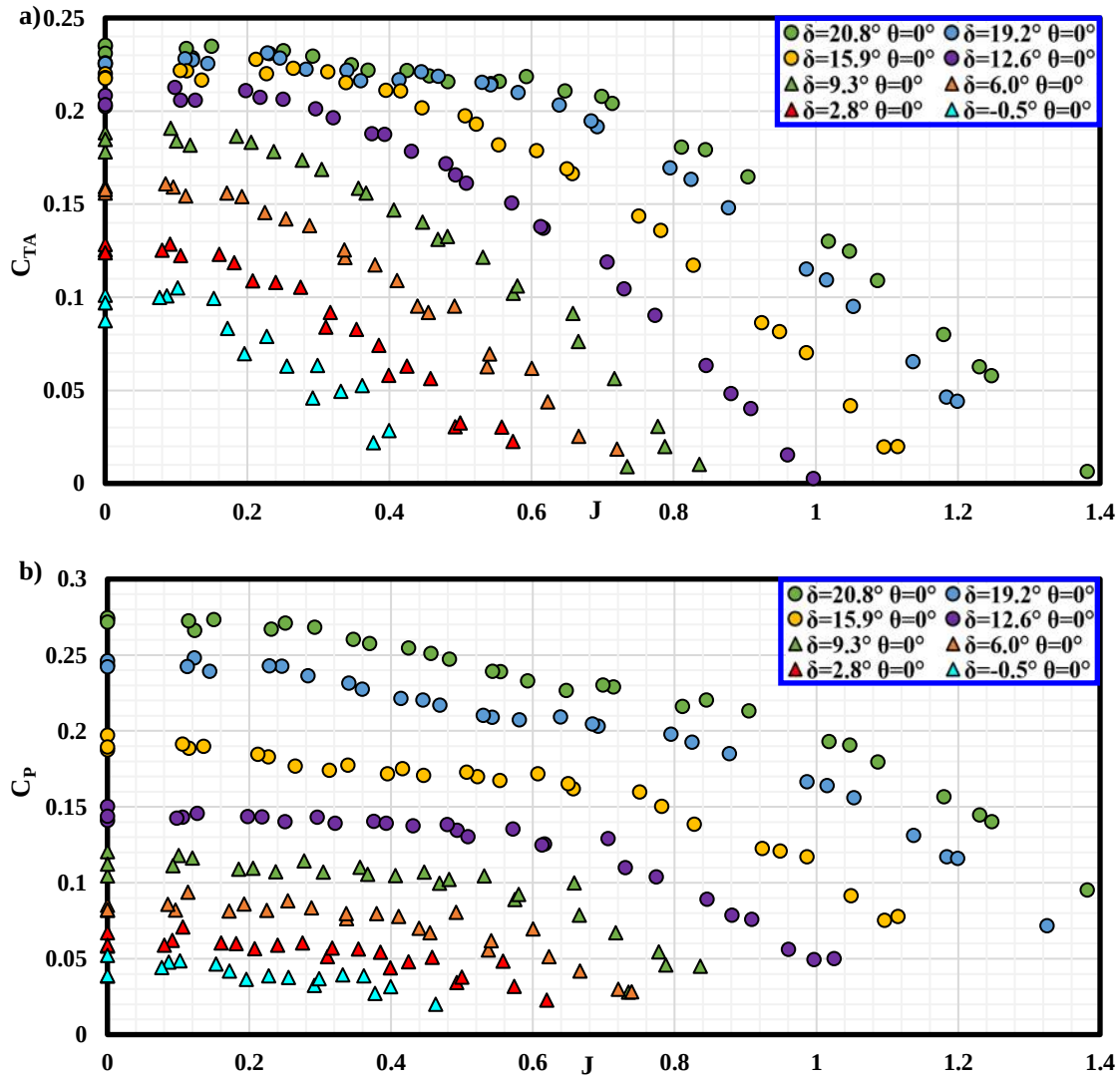


Figure 7. a) C_{TA} and b) C_P vs. J at various δ at $\theta = 0^\circ$.

An alternative to FOM is to fix the input power from the power supply, while varying the collective setting and the advance ratio, to see changes in thrust. This is shown in Figure 8. The propeller rotational speed is adjusted as δ and J are varied, to keep the power supply reading approximately at a constant 1000W. In hover, the blade pitch angle between the factory setting ($\delta \sim 0^\circ$) and $\delta = 7.7^\circ$ shows the highest thrust generated; the peak, if any,

is shallow. At $\delta > 9.3$, a lower available axial thrust is measured, which agrees with the results in Figure 6. At $J < 0.3$, the thrust generated at $\delta \sim 0^\circ$ falls significantly with the increment in J , while the thrust generated at $\delta > 14.3^\circ$ remains almost constant and underperforms the lower δ cases. There is an envelope of maximum thrust vs. advance ratio, with this maximum falling as J increases, no matter what the blade collective setting is. For example, at $J = 0.4$, the highest available thrust is at $\delta \sim 11.0^\circ$. At $J > 0.7$, the propeller generates its highest possible thrust at the maximum available collective setting ($\delta \sim 17.6^\circ$). To fly efficiently at a higher advance ratio, the mechanism would have to be revised to afford even higher δ beyond the factory setting.

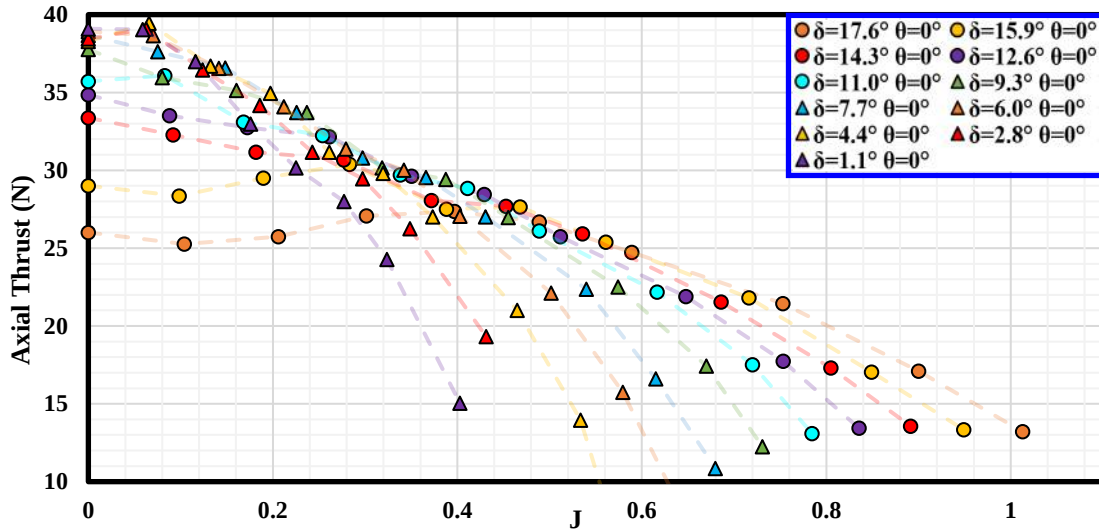


Figure 8. Axial Thrust vs. J at various δ at $\theta = 0^\circ$ and constant power (1000 W).

The more usual performance metric in forward flight is the propeller efficiency, η . The calculated η for all cases in Figure 8 is shown in Figure 9. As expected, lower δ has a higher efficiency near hover conditions. At $J < 0.4$, the $\delta \sim 6.0^\circ$ case was able to maintain a higher η , and so on, in agreement with Figure 8.

We can consider a normalized thrust coefficient for a variable pitch propeller as:

$$C_{TA(\delta,J) \text{ Norm}} = \frac{C_{TA(\delta,J)}}{C_{TA(\delta,J=0) \text{ Ideal}}} \quad (2)$$

$C_{TA(\delta,J) \text{ Norm}}$ is the ratio between the thrust coefficient at a blade pitch angle δ and advance ratio J , to the hover-reference ($C_{TA(\delta,J=0) \text{ Ideal}}$) at the same blade pitch angle δ in Equation 1.

From classical blade element theory [13], when the propeller blade experiences a higher effective angle of attack, a higher thrust is generated, and thus a higher C_{TA} . The propeller blade experiences a lower blade effective angle of attack at a higher advance ratio at the same δ . Here we normalize the advance ratio as:

$$J_{\delta \text{ Norm}} = J_{\delta} * \frac{C_{TA(\delta=0^\circ, J=0)}}{C_{TA(\delta,J=0) \text{ Ideal}}} \quad (3)$$

Where $C_{TA(\delta=0^\circ, J=0)}$ is the hover thrust coefficient at $\delta = 0^\circ$.

Figure 10 replots the results of Figure 7.a in terms of $C_{TA(\delta,J) \text{ Norm}}$ and $J_{\delta \text{ Norm}}$. A collapse of the curve between different δ is observed at $J_{\delta \text{ Norm}} > 0.25$. Below this threshold, there is still some data collapse, but limited to $\delta < 12.6^\circ$, which is to say, below stall.

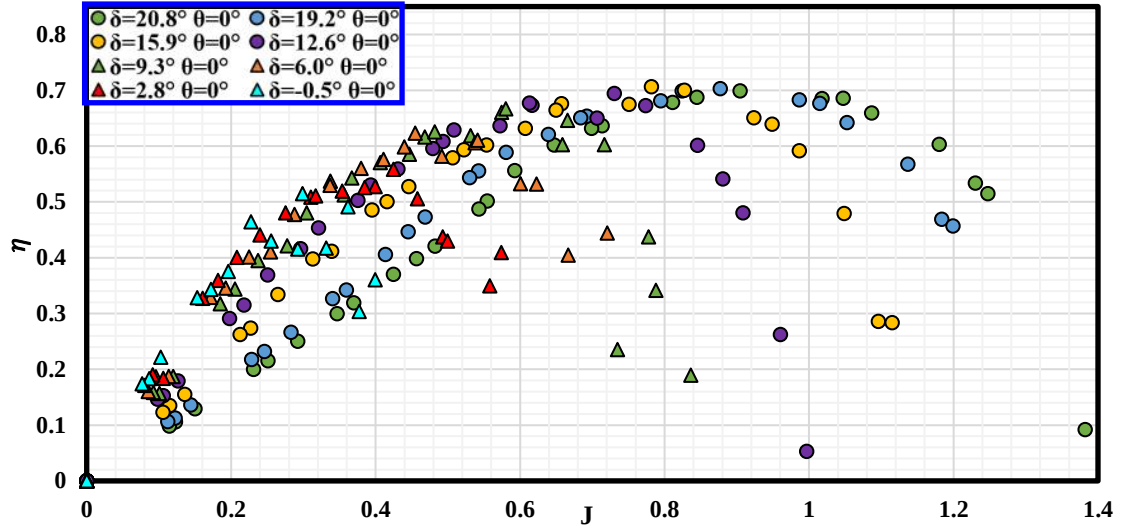


Figure 9. Propeller Efficiency (η) vs. J at various δ at $\theta = 0^\circ$.

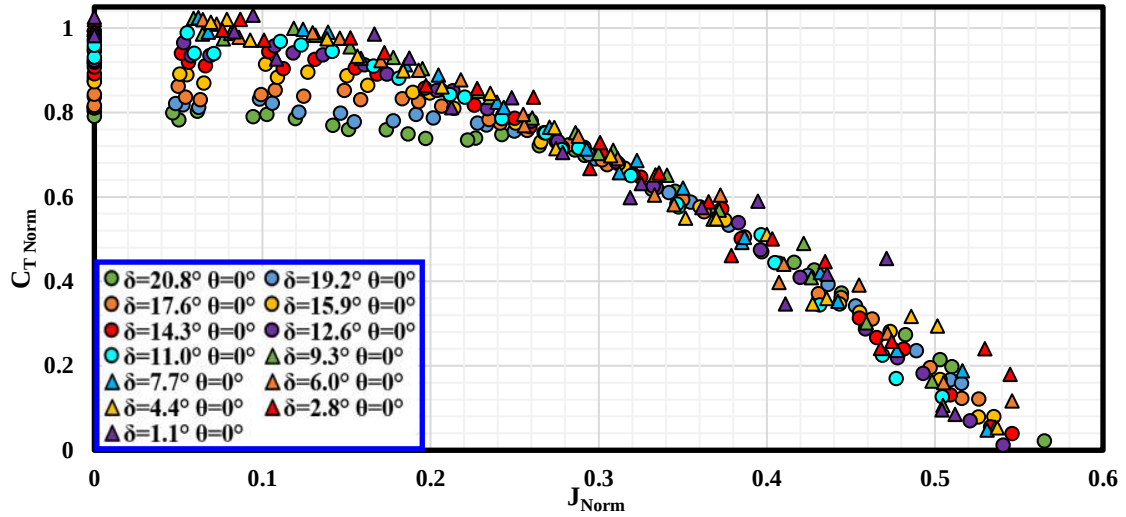


Figure 10. $C_{TA}(\delta J) Norm$ vs. $J_{\delta Norm}$ at various δ at $\theta = 0^\circ$.

Consider the ‘factory default’ condition at $\delta = 0^\circ$, where:

$$C_{TA(\delta=0^\circ, J=0)} = C_{TA(\delta=0^\circ, J=0) Ideal} \quad \text{and} \quad J_{\delta Norm} = J_{\delta=0^\circ} * \frac{C_{TA(\delta=0^\circ, J=0)}}{C_{TA(\delta, J=0) Ideal}} = J_{\delta=0^\circ} \quad (4)$$

Therefore, the collapse curve (or ‘master curve’) simply represents the $C_{TA(\delta=0^\circ, J) Norm}$ vs. $J_{\delta=0^\circ}$, which is the hover normalized C_{TA} for the variable pitch propeller at $\delta = 0^\circ$.

On the other hand, at $\delta > 12.6^\circ$, $C_{TA(\delta, J) Norm}$ remains almost the same with the increase in $J_{\delta Norm}$, until the curve collapses with the ‘master curve’. This is a result of the blade stall at small J , where the measured C_{TA} is lower than the ideal C_{TA} . When the $C_{TA(\delta, J) Norm}$ vs. $J_{\delta Norm}$ curve merge and collapses with the ‘master curve’, the blade evidently recovers from stall and the performance is thus predictable at lower δ cases.

To predict the variable pitch propeller C_{TA} at different δ and J at $\theta = 0^\circ$, two sets of data are required. The first set is C_{TA} vs. J at $\theta = 0^\circ$ and $\delta = 0^\circ$, which can be obtained from the manufacturer. Using this data, the ‘master curve’ can be obtained using Equation 2. The second set is C_{TA} vs. δ at $J = 0$, which can be

measured without a wind tunnel experiment. This provides the relationship between C_{TA} and δ (Equation 1) and the δ where blade stall occurs in hover. The step-by-step of the proposed prediction model is shown below.

- Step 1: Obtain C_{TA} vs J for a propeller of interest at $\theta = 0^\circ$. Divide $C_{TA}(J)$ by $C_{TA}(J = 0)$. This normalized C_{TA} will serve as a C_{TANorm} and J will serve as a $J_{\delta Norm}$, which is the ‘master curve’.
- Step 2: Obtain the data for C_{TA} vs. δ for the propeller of interest at $J = 0$ and apply a linear curve fit to obtain the ideal thrust coefficient at $J = 0$, $[C_{TA}(\delta_x)_{Ideal}]_{J=0}$ as a function of the slope ($d_{C_{TA}}/d_{\delta}$), and the intercept $[C_{TA}(\delta=0^\circ)]_{J=0}$. From the data, also find the δ at which the blade stalls (δ_{stall}).

$$[C_{TA}(\delta_x)_{Ideal}]_{J=0} = \left[\left(\frac{d_{C_{TA}}}{d_{\delta}} \right) * \delta_x + C_{TA \delta=0^\circ} \right]_{J=0}$$

- Step 3: Using $J_{\delta Norm}$ from Step 1 and the expression of $[C_{TA}(\delta_x)_{Ideal}]_{J=0}$ from step 2, calculate J_{δ_x} for the blade pitch angle of interest, δ_x , using Equation 3, where:

$$J_{\delta_x} = J_{\delta Norm} * \left[\frac{C_{TA}(\delta_x)_{Ideal}}{C_{TA}(\delta=0^\circ)} \right]_{J=0} \quad (5)$$

- Step 4: Calculate the ideal C_{TA} for δ_x at J_{δ_x} , $[C_{TA}(\delta_x)_{Ideal}]_{J_{\delta_x}}$ using C_{TANorm} from Step 1, $[C_{TA}(\delta_x)_{Ideal}]_{J=0}$ from Step 2 and Equation 2, where:

$$[C_{TA}(\delta_x)_{Ideal}]_{J_{\delta_x}} = C_{TANorm} * [C_{TA}(\delta_x)_{Ideal}]_{J=0} \quad (6)$$

- Step 5: Determine the predicted C_{TA} for δ_x at J_{δ_x} where:

$$[C_{TA}(\delta_x)]_{J_{\delta_x}} = \min \left[[C_{TA}(\delta_x)_{Ideal}]_{J_{\delta_x}}, [C_{TA}(\delta_x)]_{J=0} \right] \quad (7)$$

Equation 7 indicates that when the blade stall does not occur in hover, where $[C_{TA}(\delta_x)]_{J=0}$ is equal to $[C_{TA}(\delta_x)_{Ideal}]_{J=0}$ and $[C_{TA}(\delta_x)_{Ideal}]_{J_{\delta_x}}$ is small than $[C_{TA}(\delta_x)]_{J=0}$ at $J > 0$ (as seen in Figure 10), $[C_{TA}(\delta_x)_{Ideal}]_{J_{\delta_x}}$ from Equation 6 represents the predicted propeller performance at δ and J . However, when blade stall occurs in hover, where $[C_{TA}(\delta_x)]_{J=0}$ is small than $[C_{TA}(\delta_x)_{Ideal}]_{J_{\delta_x}}$, $[C_{TA}(\delta_x)]_{J_{\delta_x}}$ will remain at the same value as the propeller performance in hover $[C_{TA}(\delta_x)]_{J=0}$ at $J > 0$ until the blade recover from stall. This occurs at the advanced ratio where $[C_{TA}(\delta_x)]_{J=0}$ is equal to $[C_{TA}(\delta_x)_{Ideal}]_{J_{\delta_x}}$, and the propeller starts to perform similar to the ideal condition as smaller δ cases where the blade stall does not

occur. In this case, $[C_{TA}(\delta_x)]_{J\delta_x}$ can be represented by $[C_{TA}(\delta_x)_{Ideal}]_{J\delta_x}$ as it is less than $[C_{TA}(\delta_x)]_{J=0}$ beyond this very advanced ratio. This trend also agrees with the observation in Figure 10.

The proposed model provides a fast prediction of propeller axial thrust in forward flight at various collective pitch settings, especially when using the off-the-shelf propeller blade and the propeller geometry is not available.

C. Forward Flight at $\theta \neq 0^\circ$

Figure 11 shows C_{TA} and C_P vs. J under the same δ as in Figure 7 at various incidence angles θ . Each color represents the data at the same θ . A very slight increase in C_{TA} and C_P are observed with increasing θ at the same J , which is consistent with the literature [4,5, 9-11] and our previous study [14]. Larger incidence angles would be needed to evince any substantial trends in thrust. A prediction model for propeller axial thrust at a higher incidence angle is also proposed in the literature using an equivalent axial advance ratio [4,11,14] and a combination of propeller-like and propeller-wing-like components [15]. This also becomes more relevant with a higher incidence angle.

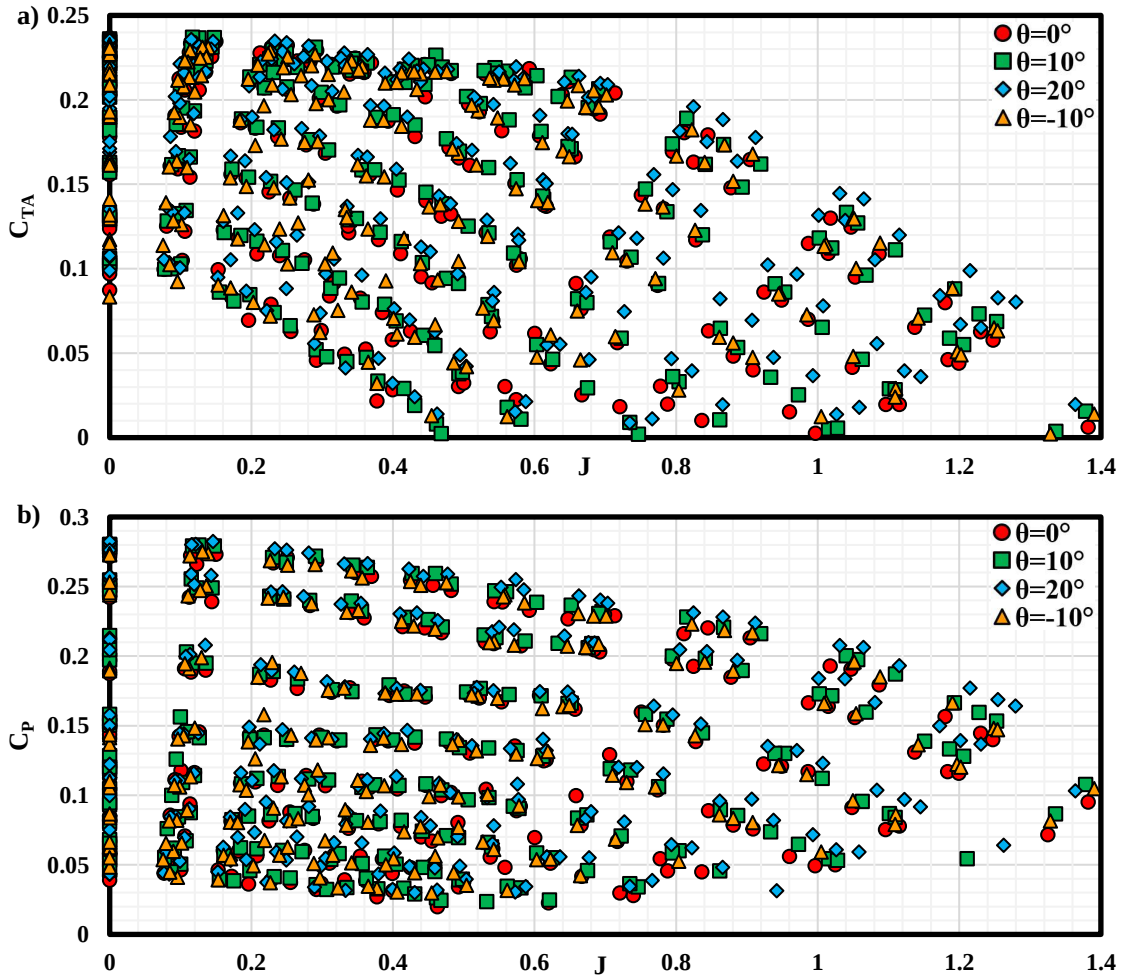


Figure 11. a) C_{TA} and b) C_P vs. J at Various δ in Forward Flight at Various θ .

Figure 12 shows the propeller pitching moment (C_m) vs. J at various θ , with a positive C_m representing a pitch-up moment and vice versa. Similar to Figure 11, each color mark presents the data at the same θ . A strong grouping of the data is observed at the same θ , indicating a similarity of the propeller pitching moment at different

δ . In general, at $\theta < 0^\circ$, a pitch-up moment is produced, while a pitch-down moment occurs at $\theta > 0^\circ$, indicating the tendency of the propeller to return to axial flight condition in forward flight. With increase in J , the magnitude of C_m also increases, evidently as a result of azimuthally uneven blade loading. This is again consistent with the literature [4,11] and with our previous study on fixed-pitch propellers [14].

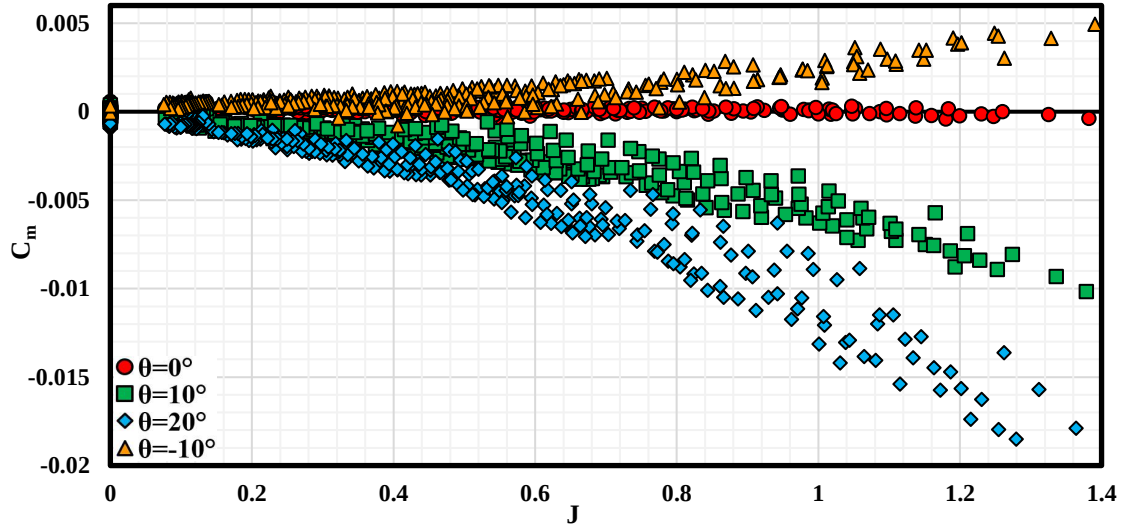


Figure 12. C_m vs. J at Various δ in Forward Flight at Various θ .

Figure 13.a shows C_m vs. J for selected cases at $\theta = 20^\circ$. The magnitude of C_m increases with J and the trend remains even at $C_{TA} < 0$. A good collapse is observed between almost all δ cases, indicating a strong relationship C_m and θ as well as the potential predictability of the propeller pitching moment under various δ . C_m does deviate from the main curve at higher J , after the propeller reaches its maximum η (as seen in Figure 14) and the effective blade angle of attack starts falling substantially. By inspection, one can take as a threshold, the condition for advance ratio beyond that of peak efficiency, where efficiency has fallen to 80% of peak. That is, take the advance ratio for peak efficiency; then examine data points for higher J , as efficiency falls. Eventually, the efficiency falls to 80% of the peak. Then, expunge all data points of higher J . The result is Figure 13b, which evinces a tighter collapse of data points than is Figure 13a.

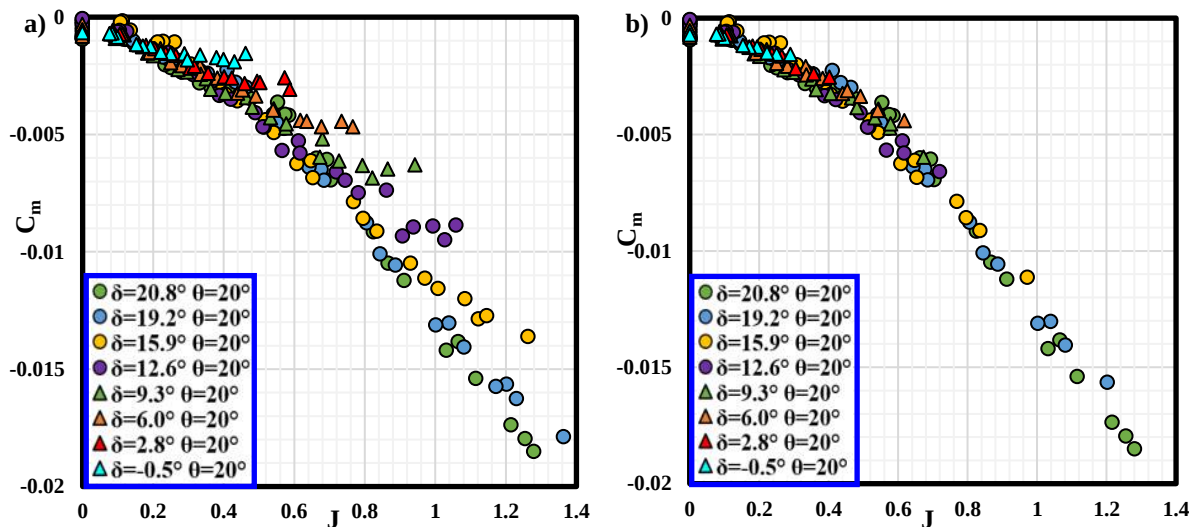


Figure 13. C_m vs. J at Various δ at $\theta = 20^\circ$; a) Full and b) Parsed data where the data points past $0.8 \eta_{max}$ are removed.

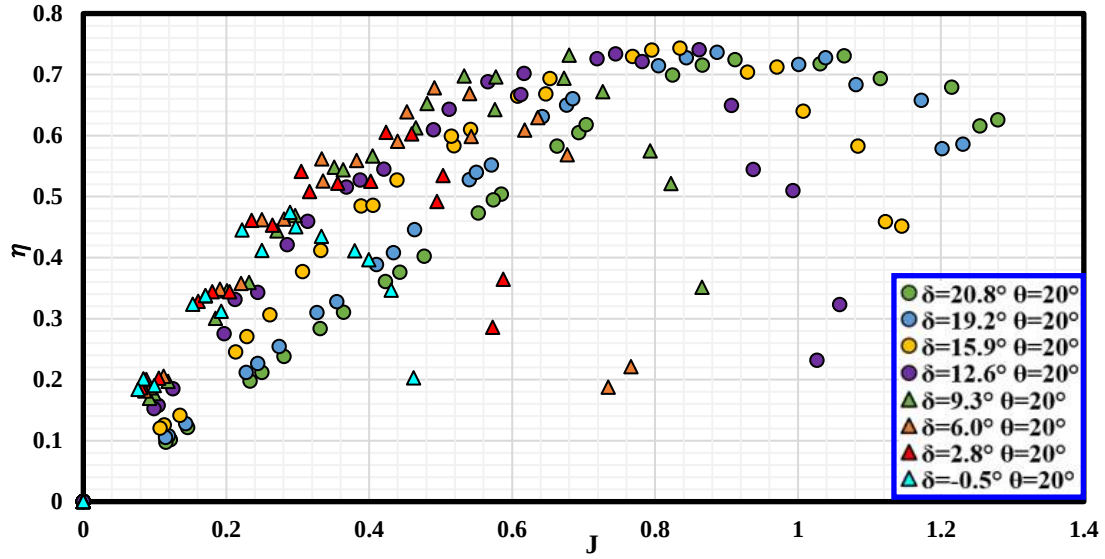


Figure 14. Propeller Efficiency (η) vs. J at various δ at $\theta = 20^\circ$.

Figure 15 shows the average calculated stability derivative $\frac{dC_m}{d\theta}$ vs. δ at different J ; $\frac{dC_m}{d\theta}$ increases with J and decreases with δ . If we expunge the data points for advance ratio beyond that for which efficiency has fallen more than 20% from its peak value, as was done in Figure 13.b, and recalculate $\frac{dC_m}{d\theta}$ for the remaining (parsed) data points, then $\frac{dC_m}{d\theta}$ becomes essentially constant with collective pitch, and only varies with advance ratio.

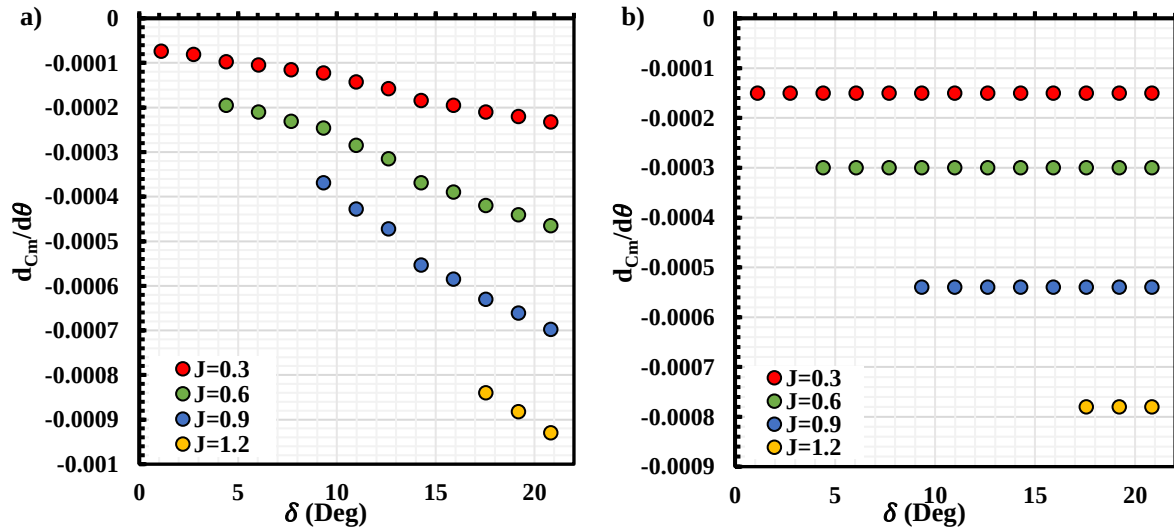


Figure 15. $\frac{dC_m}{d\theta}$ vs. δ at Various J for a) full and b) parsed data.

The propeller yawing moment (C_n) vs. J and θ is shown in Figure 16. A positive C_n is the leftward yawing moment of the propeller disk as a result of the asymmetric blade loading at $\theta \neq 0^\circ$. To be clear, θ is also a pitching angle, in the sense of the aircraft rigid-body dynamics. Pitching moment with respect to θ is by

symmetry a yawing moment with respect to the sideslip. The sideslip is identically zero in this study, but the yawing moment is measured in all cases anyway. Figure 16 then considers the less common stability derivative of the yawing moment with respect to the pitching angle; that is, with respect to θ . When compared to the results for C_m in Figure 12, more scatter is observed. This is more clearly seen in the C_n data limited to just $\theta = 20^\circ$ in Figure 17. Higher δ means higher C_n for a given J . This can also be seen in Figure 18, where the average calculated $\frac{dC_n}{d\theta}$ is plotted against δ and the results show a good linear relationship between $\frac{dC_n}{d\theta}$ and δ .

Different from the result of C_m , C_n vs. J remains almost linear across all J . Therefore, by normalizing $\frac{dC_n}{d\theta}$ by the advance ratio J , yields $\left(\frac{dC_n}{d\theta}\right)/d_J$, and the results Figure 18.a collapse to the single curve in Figure 18.b. The relationship between $\left(\frac{dC_n}{d\theta}\right)/d_J$ and δ is almost linear, indicating good predictability of the propeller yawing moment at higher θ under different δ .

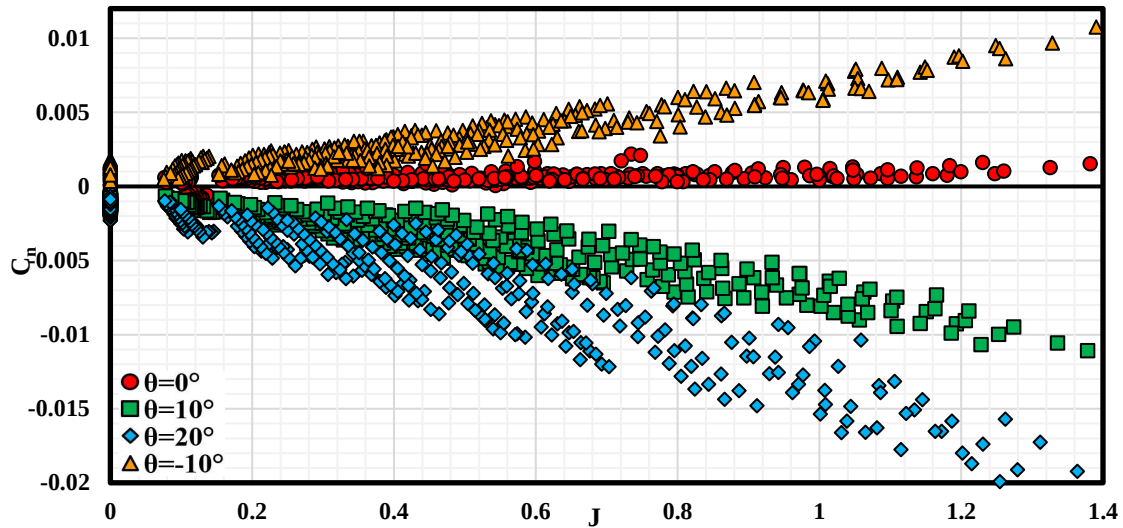


Figure 16. C_n vs. J at various δ and θ .

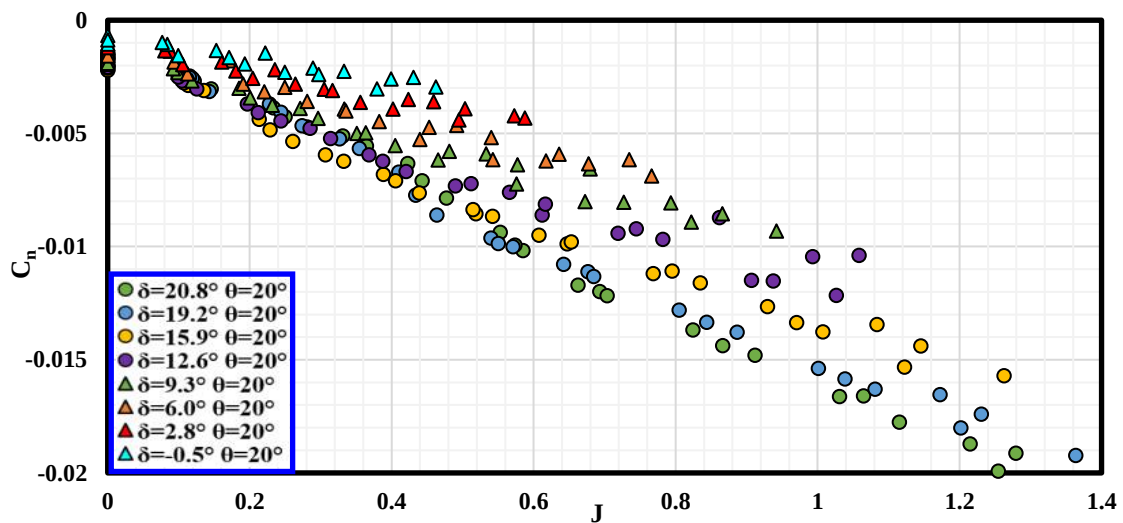


Figure 17. C_n vs. J at various δ at $\theta = 20^\circ$.

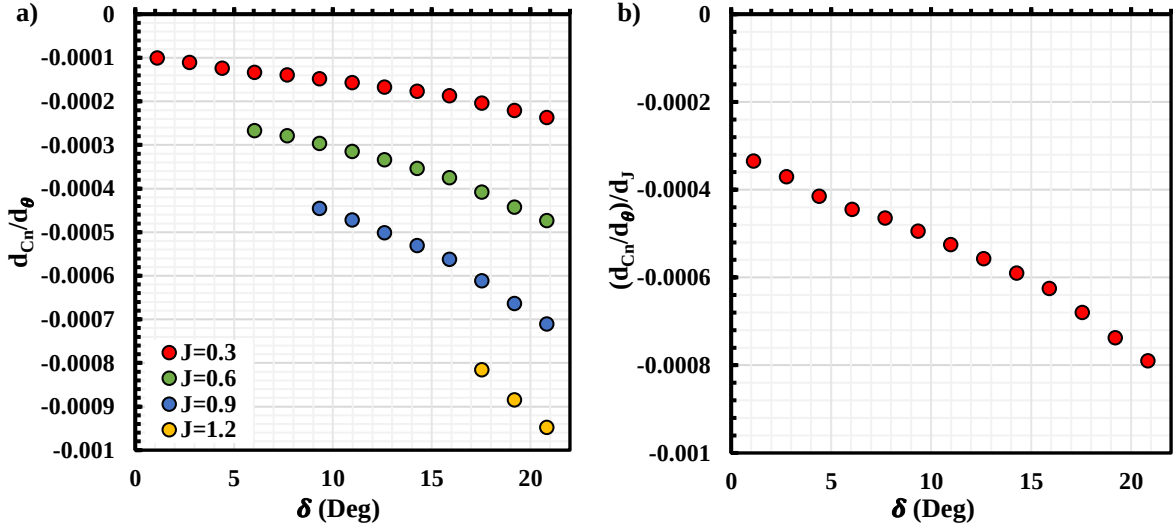


Figure 18. a) $\frac{dC_n}{d\theta}$ at various J and b) $\left(\frac{dC_n}{d\theta}\right)/dJ$ at various δ .

V. Conclusions and Future Work

The aerodynamic performance of a small-scale variable collective pitch propeller is experimentally investigated at incidence angles between $-10^\circ \leq \theta \leq 20^\circ$. In hover, a linear relationship of thrust and collective pitch is observed between $-8^\circ \leq \delta \leq 12^\circ$, with stall above this range.

Performance in axial forward flight follows the general trend observed in the literature [11], where a higher δ leads to a better propeller performance at higher speed, and vice versa. Using the ideal relationship between C_T and δ in hover, the measured C_T and J in axial forward flight under different δ can be normalized, resulting in a curve collapse, except for the higher δ at smaller J where blade stall is experienced by the propeller. This indicates a good thrust predictability of the variable pitch propeller in axial forward flight.

At nearly but not quite axial flight condition ($\theta \neq 0^\circ$), a higher propeller thrust, power consumption, pitching, and yawing moment are observed, which is consistent with the literature [4,9,11] and our previous study [14]. C_m shows a strong independence on δ until reaching an advanced ratio threshold, while a higher C_n is observed at higher δ . Good predictability of C_m and C_n at various δ , θ and J is also concluded within the incidence angle range examined in the current study.

Future studies will extend the test matrix to higher incidence angles, near and beyond edgewise flight ($\theta > 90^\circ$).

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